

Advanced Pivoting, Persistence Simulation & Cleanup

Pivoting catalog (beyond basics)

Technique	Use case
SSH -L/R/D	classic tunnel (already learned)
Socks5 + proxychains	sweep the far net
Port forwarding in heartbeat (C2)	dynamic tunnels via implant
SSH remote forward (R)	make your box reachable FROM internal ('expose kali')
Netcat relays / socat	chain hops without SSH
DNS tunneling (iodine/dns2)	when only DNS leaves
Metasploit route + portproxy	MSF-native toggle

```
# socat relay example (lab):
socat TCP-LISTEN:8888,fork TCP:10.0.0.5:80
# ssh remote forward: external into internal service
ssh -R 127.0.0.1:8443:10.0.5.10:443 user@internal-box
```

Persistence simulation & golden-ticket

Golden ticket (lab demo): DCSync krbtgt → mint fakeadmin tickets (as learned in M28) – used in emulation as 'the adversary would keep reign'

Scheduled-tasks/Run keys (simulate) – mark every persistence intent with a clearly-labeled marker

Purpose: teach blue where to look; never leave REAL persistence on prod

Cleanup = the engagement's last 20% (don't skip)

Checklist per engagement:

- restore modified configs/registry/ADS
- remove created users, group memberships, scheduled tasks/services
- delete uploaded files, download cradles, temp artifacts
- clean added firewall rules, DNS entries
- revoke/delete test credentials & tokens
- final: re-run your scan = system matches baseline

Why cleanup is non-negotiable

Leaving backdoors = illegal in most scopes (and dangerous)

It's the difference between 'a test' and 'a crime with your name on it'

Blue handout: 'here is exactly what we touched, restore it yourself, done'

The pivot chain is how engagements reach objectives; the cleanup is how professionals stay licensed. Both matter equally.

[← Adversary Emulation: Build a...](#)

[Reporting & Blue-Team Collabor...](#)